

WPR - 5

AMITY SCHOOL OF ENGINEERING & TECHNOLOGY
B.Tech (CSE - AIML) VII Semester

Project Area -- Generative AI & AI Safety (Trustworthy AI / MLOps)

Students Weekly Progress Report (WPR — Week 5) For Odd Semester of session 2026-2027

To be filled by Students	Group No. 50
Students Name	1. Shourya Solanki 2. Rachit Ryan Chug 3. Dhruv Raj Singh
Roll no.	1. 569 2. 166 3. 191
Enrollment no.	1. A2305223569 2. A2305223166 3. A2305223191
Project Title finalized, if Yes, give name, if NO, give reason	Yes. Title: "GuardShield AI: Real-Time Sidecar Proxy for LLM Hallucination Detection & Prompt Injection Defense"
Synopsis submitted	Yes (Approved for Group No. 50)
Literature review	Applied to latency benchmarking and inference optimization. Verified DeBERTa-v3 sub-25ms pre-scan latency constraint on test split.
Technical & Economical Feasibility	Feasible. Python 3.11+, PyTorch CUDA (RTX 3050 GPU), DeBERTa-v3 SLM (86M params), standalone inference engine. Zero monetary cost (₹0).
Bill of Material	NA
Project Progress Schedule (PERT Chart)	- W1-W3: Setup, open-source dataset ingestion & GuardShield-Bench-v1 synthetic creation (Completed) - W4-W7: DeBERTa-v3 model fine-tuning & evaluation pipeline (In Progress / Week 5) - W8-W12: Token logit entropy algorithm, benchmark evaluation & IEEE paper draft
Design of critical components	1. Pre-Scan Model Evaluation Script (`evaluate_prescan.py`) 2. Pre-Scan Security Filter Engine (`prescan_filter.py`) 3. Quantitative Accuracy & Latency Metrics Pipeline
Fabrication work (give %)	0% (Software Project)
Experimental work (give %)	55% (Pre-scan evaluation executed on 41 test samples, accuracy and F1 metrics calculated, sub-25ms inference latency verified on CUDA GPU, standalone filter engine implemented)
Result and Analysis	Evaluated fine-tuned DeBERTa-v3 on test.json; confirmed 87.8% Accuracy and sub-20ms average inference latency on CUDA GPU, meeting real-time sidecar requirements.
Report writing	WPR Week 5 drafted.

Signature of students	1. Shourya Solanki 3. Dhruv Raj Singh	2. Rachit Ryan Chug
-----------------------	--	---------------------

Solanki *Rachit* *Dhruv*

Work done in this week (Week 5):

1. Model Evaluation Pipeline Implementation: Developed `evaluate\_prescan.py` to benchmark the fine-tuned `microsoft/deberta-v3-small` classifier across 41 test samples from `test.json`.
2. Performance & Metrics Quantification: Measured Classification Accuracy, Precision, Recall, and F1-Score on prompt injection and adversarial test prompts.
3. Latency Benchmarking on CUDA GPU: Benchmarked per-request inference overhead on local NVIDIA GeForce RTX 3050 GPU, confirming sub-20ms latency (within the sub-25ms real-time constraint).
4. Standalone Security Filter Module: Implemented `src/prescan\_filter.py` providing a production-ready interface returning safety decision (ALLOW/BLOCK), risk score, and execution latency in milliseconds.

To be filled by Guide (strike off whichever is not applicable)	
Performance of students is satisfactory	
Performance of students is unsatisfactory	
A warning to be issued to student(s) (Name)	
Student was not well (Name)	
Date 26/08/2026	Signature of Guide